

MELAASSIST

AI-Powered Safety and Navigation Assistant for Large-Scale Religious Gatherings

A Project Report

submitted in partial fulfillment of the requirements for internship

under

Work Based Learning (WBL)

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Academic Session 2025–2026

Abstract

Large-scale religious gatherings such as melas, festivals, fairs, and pilgrimage events attract thousands of visitors within a limited geographical area. Managing visitor safety, emergency response, navigation, and crowd movement in such environments remains a major challenge due to the absence of intelligent digital infrastructure.

This project proposes MelaAssist, an AI-powered integrated platform combining crowd monitoring, anomaly detection, navigation assistance, and emergency response. The system employs computer vision techniques using YOLOv8, CSRNet, and ResNet-50 to monitor crowd density and detect unusual crowd behavior. Simultaneously, a GPS-enabled navigation and SOS module assists visitors in locating essential services such as medical camps, police stations, exits, and information centers.

The proposed architecture creates a closed-loop ecosystem where crowd intelligence directly influences navigation recommendations and emergency alerts.

Keywords: Crowd Monitoring, YOLOv8, CSRNet, ResNet-50, Navigation System, SOS Alert, Crowd Density Estimation.

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Chapter 1

Introduction

Religious gatherings and festivals are an integral part of cultural traditions across India and many parts of the world. Events such as Kumbh Mela, Parshuram Kund Mela, Durga Puja celebrations, and temple festivals attract large crowds, often exceeding the capacity of available infrastructure. Managing such gatherings requires ensuring visitor safety, emergency response readiness, and efficient crowd movement. Traditional management methods rely heavily on manual monitoring and static signboards, which become ineffective as crowd density increases. Recent advances in Artificial Intelligence, Computer Vision, and Geospatial Technologies provide opportunities to develop intelligent systems capable of assisting both visitors and event organizers. MelaAssist leverages these technologies to create a unified platform that provides:

- **Real-time crowd monitoring :** The real-time crowd monitoring module serves as the foundation of the MelaAssist system. It continuously processes live CCTV or surveillance camera feeds deployed across the event venue to monitor crowd activities. The objective of this module is to provide event organizers with a comprehensive understanding of crowd movement patterns and occupancy levels across different zones. By leveraging computer vision and deep learning techniques, the system can automatically identify and track individuals in crowded environments, reducing dependence on manual monitoring and enabling timely decision-making during large-scale public gatherings.
- **Crowd density estimation :** Crowd density estimation is a critical component of the proposed system, aimed at measuring the concentration of people within different areas of the venue. After detecting individuals using a person detection model, the system employs CSRNet to generate density maps and estimate crowd counts in real time. These density estimates help organizers identify congested zones and potential bottlenecks before they become hazardous. The generated density information is also integrated with the navigation module to guide visitors through safer and less crowded routes.

- **Anomaly detection:**The anomaly detection module is designed to identify unusual or potentially dangerous crowd behaviors that may indicate safety risks. A fine-tuned ResNet-50 model analyzes crowd movement patterns captured from video streams and classifies them into categories such as normal movement, dense/static crowds, sudden surges, or dispersal events. By detecting abnormal crowd behavior at an early stage, the system can alert organizers and security personnel, allowing preventive actions to be taken before situations escalate into emergencies such as stampedes or crowd panic.
- **Emergency SOS support:** The emergency SOS support system provides visitors with a quick and reliable method for requesting assistance during emergencies. When a visitor activates the SOS feature through the mobile or web application, their current GPS location is automatically transmitted to the backend server and forwarded to the nearest available responder. This eliminates the need for users to manually communicate their location, thereby reducing response time and improving the effectiveness of emergency services. The SOS module plays a crucial role in enhancing safety and ensuring rapid intervention during critical situations.
- **GPS-based navigation:**The GPS-based navigation module assists visitors in locating important services and destinations within the event premises. Using the visitor's real-time location, the system identifies the nearest facilities such as medical camps, police stations, washrooms, information centers, and emergency exits. Navigation instructions are generated dynamically based on current crowd conditions, enabling visitors to reach their destinations efficiently while avoiding highly congested areas. This feature significantly improves accessibility and overall visitor experience during large gatherings.
- **Organizer dashboard:**The organizer dashboard acts as a centralized control interface for event management authorities. It provides real-time visualization of crowd density levels, anomaly alerts, emergency requests, and overall venue status. Information from the crowd monitoring and navigation modules is aggregated and displayed in an intuitive format, enabling organizers to monitor the event continuously and make informed decisions. The dashboard serves as the primary tool for crowd management, risk assessment, and emergency coordination, contributing to a safer and more organized event environment.

Chapter 2

Literature Review

The increasing occurrence of large-scale public gatherings has motivated researchers to develop intelligent systems for crowd monitoring, safety management, and navigation assistance. Recent advances in computer vision and deep learning have significantly improved the ability to analyze crowd behavior, estimate density, and detect abnormal activities in real time.

Shao et al. introduced CrowdHuman, a large-scale benchmark dataset specifically designed for human detection in crowded scenes with severe occlusions and overlapping individuals [1]. The dataset contains over 470,000 annotated human instances and has become a widely used benchmark for training and evaluating person detection models in dense crowd environments. The availability of such a dataset has enabled the development of robust crowd analysis systems capable of operating in challenging real-world conditions.

Li et al. proposed CSRNet, a crowd counting framework based on dilated convolutional neural networks for understanding highly congested scenes [2]. By replacing pooling layers with dilated convolutions, CSRNet effectively preserves spatial information while expanding the receptive field. Experimental results demonstrated state-of-the-art performance on several crowd counting benchmarks, making CSRNet one of the most widely adopted models for crowd density estimation in surveillance applications.

Sultani et al. introduced the UCF-Crime dataset and presented a deep learning framework for anomaly detection in surveillance videos [3]. The dataset consists of real-world videos containing abnormal events such as accidents, violence, and crowd disturbances. Their work demonstrated that deep neural networks can successfully identify unusual activities from video streams, providing an effective foundation for automated crowd behavior monitoring and public safety systems.

Jocher et al. developed YOLOv8, an advanced object detection framework that achieves real-time performance while maintaining high detection accuracy [4]. YOLOv8 incorporates architectural improvements and optimized training strategies that enable efficient detection of individuals in crowded environments. Due to its speed and accuracy, YOLOv8 has become a preferred choice for surveillance and crowd monitoring

applications where rapid processing of live video feeds is essential.

He et al. proposed ResNet-50, a deep residual neural network architecture that addresses the degradation problem in deep learning models through residual connections [5]. ResNet-50 has been widely adopted for image classification, feature extraction, and behavior recognition tasks. In crowd monitoring applications, the model can be fine-tuned to classify crowd movement patterns and identify abnormal situations such as sudden surges, panic movement, or dispersal events.

Although these studies have significantly advanced crowd monitoring and public safety technologies, most existing solutions focus on individual tasks such as person detection, crowd counting, anomaly detection, or navigation independently. There remains a lack of integrated systems that combine crowd intelligence with real-time navigation and emergency response mechanisms. The proposed MelaAssist framework aims to bridge this gap by integrating crowd density estimation, anomaly detection, GPS-based navigation, and SOS support into a unified platform for large-scale religious gatherings and public events.

Chapter 3

Research Gap

Most existing solutions focus on either crowd monitoring or navigation independently. Very few systems integrate crowd analytics, anomaly detection, navigation, and emergency response into a unified framework. MelaAssist addresses this limitation by integrating AI-based crowd intelligence with location-aware navigation and emergency support.

Chapter 4

Dataset Description

The proposed MelaAssist system utilizes three publicly available datasets to support different stages of the crowd monitoring pipeline. The CrowdHuman dataset is employed for training the YOLOv8 person detection model, as it contains densely populated scenes with significant human overlap and occlusion. For crowd density estimation, the UCF-QNRF dataset is used due to its large collection of high-resolution images featuring extremely dense crowds and accurate crowd count annotations. To facilitate anomaly and crowd behavior detection, the UCF-Crime dataset is utilized, providing real-world surveillance videos containing both normal and abnormal events such as panic situations, disturbances, and unusual crowd activities. For the navigation and SOS module, no dedicated training dataset is required; instead, sample GPS coordinates and service location data, including medical camps, security posts, and emergency exits, are used to develop and evaluate the geospatial functionalities of the system.

Chapter 5

Methodology

The proposed methodology consists of four major phases: dataset collection and preprocessing, crowd monitoring model development, navigation and SOS system implementation, and system integration and evaluation. The crowd monitoring pipeline utilizes deep learning models for person detection, crowd density estimation, and anomaly detection, while the navigation module employs GPS-based geospatial services to provide location-aware assistance and emergency support. The integration of these modules enables real-time crowd intelligence to directly influence navigation recommendations and emergency response mechanisms.

5.1 Dataset Collection and Preprocessing

The proposed system utilizes three publicly available datasets for training and evaluation. The CrowdHuman dataset is used for person detection due to its large number of densely crowded scenes with significant human occlusions. The UCF-QNRF dataset is employed for crowd density estimation because it contains high-resolution images with extremely dense crowds and accurate annotations. For anomaly detection, the UCF-Crime dataset provides surveillance videos containing both normal and abnormal events. During preprocessing, images and video frames are resized to a uniform resolution, annotations are standardized, and normalization techniques are applied to improve model training performance. The datasets are then divided into training, validation, and testing sets to ensure reliable evaluation.

5.2 Crowd Monitoring Module

The crowd monitoring module begins with live CCTV or surveillance camera feeds from the event venue. Video frames are first processed using the YOLOv8 object detection model to identify individuals in crowded scenes. The detected outputs are then passed

to CSRNet, which generates density maps and estimates crowd density across different regions. Subsequently, a fine-tuned ResNet-50 model analyzes crowd movement patterns and classifies crowd behavior into categories such as normal, dense/static, surge, or dispersal. The resulting crowd intelligence information is displayed on the administrator dashboard for real-time monitoring and decision-making.

5.3 Navigation and SOS Module

The navigation and SOS module is designed to assist visitors during large public gatherings. A mobile or web application continuously acquires the user’s GPS location and communicates with a FastAPI-based backend server. The backend accesses a geospatial database containing the coordinates of important service locations, including medical camps, police stations, emergency exits, and information centers. Based on the user’s current position, the system recommends the nearest available services and generates navigation guidance. In emergency situations, visitors can trigger an SOS alert, which automatically transmits their location to the nearest responder for immediate assistance.

5.4 System Integration and Evaluation

The final phase focuses on integrating the crowd monitoring and navigation modules into a unified platform. Crowd density estimates and anomaly alerts generated by the computer vision pipeline are transmitted to the backend server in real time. The navigation module utilizes this information to avoid congested or unsafe zones while generating route recommendations. System performance is evaluated using metrics such as Precision, Recall, F1-Score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and response time. The integrated system is expected to improve crowd safety, enhance visitor navigation, and enable proactive emergency management during large-scale religious gatherings.

Chapter 6

System Architecture

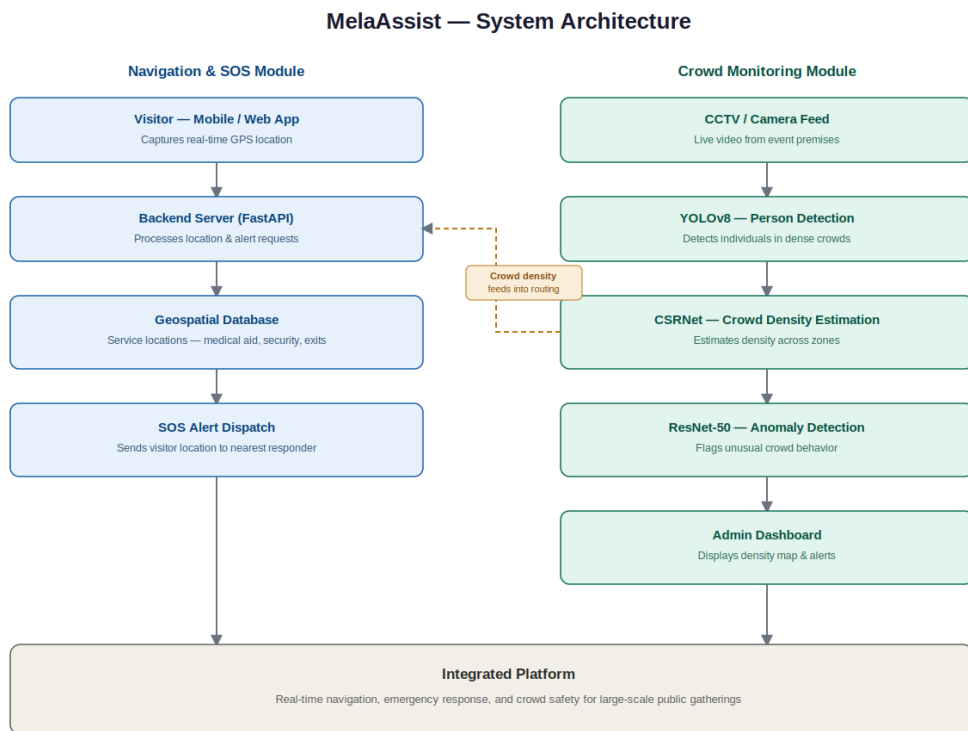


Figure 6.1: MelaAssist System Architecture

Chapter 7

Expected Results

The proposed MelaAssist system is expected to provide an integrated solution for crowd monitoring, visitor navigation, and emergency response during large-scale public gatherings. The YOLOv8-based person detection model is expected to accurately identify individuals in dense crowd environments, while CSRNet should generate reliable crowd density maps capable of highlighting congested zones in real time. The ResNet-50 anomaly detection model is anticipated to classify crowd behavior effectively and identify potentially unsafe situations such as sudden crowd surges, abnormal movement patterns, or overcrowding before they escalate into emergencies.

In addition to crowd monitoring, the GPS-based navigation module is expected to assist visitors in locating essential services such as medical camps, police stations, information centers, washrooms, and emergency exits based on their real-time location. The SOS module is expected to reduce emergency response time by automatically transmitting the visitor's location to the nearest available responder. By integrating crowd intelligence with navigation services, the system should dynamically recommend safer routes and redirect visitors away from highly congested or high-risk areas.

Overall, the proposed framework is expected to enhance visitor safety, improve crowd management efficiency, and provide organizers with real-time situational awareness through a centralized dashboard. The successful implementation of MelaAssist will demonstrate the feasibility of combining computer vision, geospatial technologies, and emergency response mechanisms into a unified platform for smart event management. The developed system can further serve as a scalable solution for religious gatherings, festivals, sports events, concerts, exhibitions, and other large public gatherings.

Chapter 8

Future Work

Although the proposed MelaAssist framework provides an integrated solution for crowd monitoring, navigation, and emergency response, several enhancements can be incorporated in future versions to improve its functionality and scalability. One possible extension is the deployment of the system on edge devices and embedded platforms, allowing real-time processing of video streams with reduced dependence on cloud infrastructure and improved response times.

Future work will also focus on incorporating advanced Vision-Language Models (VLMs) such as Qwen-VL or LLaVA to enable natural language interaction with surveillance systems. Such models can help organizers query crowd conditions using textual commands and generate automatic situation reports from live video feeds. In addition, multimodal AI systems can be integrated to combine visual information, GPS data, and textual reports for enhanced situational awareness.

The crowd monitoring module can be further improved through the use of Graph Neural Networks (GNNs) and temporal forecasting models to predict crowd movement patterns before congestion occurs. This would enable proactive crowd management by forecasting high-density regions and recommending preventive actions to organizers. Similarly, reinforcement learning-based route optimization techniques can be explored to generate adaptive navigation paths that continuously respond to changing crowd conditions.

Another important enhancement involves the integration of drone-based surveillance systems to complement fixed CCTV cameras. Aerial monitoring can provide a broader view of the venue and improve the detection of crowd surges or bottlenecks in areas that are not adequately covered by ground cameras. The navigation system can also be expanded to support offline maps, multilingual assistance, and voice-based guidance, making it more accessible to visitors from diverse backgrounds.

Furthermore, future versions of MelaAssist can incorporate mobile application deployment, real-time push notifications, and direct integration with emergency services and local authorities. Large-scale pilot deployments at events such as Parshuram Kund Mela, Kumbh Mela, sports tournaments, and cultural festivals will help evaluate the system un-

der real-world conditions and facilitate further improvements. These enhancements will transform MelaAssist into a comprehensive smart-event management platform capable of supporting safer, more efficient, and technology-driven public gatherings.

Chapter 9

Conclusion

Large-scale public gatherings such as religious festivals, melas, fairs, and cultural events present significant challenges in terms of crowd management, visitor navigation, and emergency response. Traditional management approaches often rely on manual monitoring and static infrastructure, which may be insufficient in highly dynamic and densely populated environments. To address these challenges, this work proposes *MelaAssist*, an AI-powered safety and navigation assistant that integrates crowd monitoring, anomaly detection, GPS-based navigation, and SOS emergency support into a unified platform.

The proposed system combines YOLOv8 for person detection, CSRNet for crowd density estimation, and ResNet-50 for crowd behavior analysis to provide real-time situational awareness to event organizers. Simultaneously, the navigation and SOS module assists visitors in locating essential services and obtaining emergency assistance based on their current location. A key contribution of this work is the integration of crowd intelligence with navigation services, enabling the system to dynamically reroute visitors away from congested or potentially unsafe areas. This closed-loop architecture transforms crowd analytics from a passive monitoring tool into an active decision-support mechanism for both organizers and attendees.

The successful implementation of *MelaAssist* has the potential to improve visitor safety, reduce emergency response time, enhance crowd management efficiency, and support data-driven decision-making during large-scale public events. Furthermore, the proposed framework establishes a foundation for future smart-event management systems that leverage advances in artificial intelligence, computer vision, and geospatial technologies. With further development and real-world deployment, *MelaAssist* can become a scalable and practical solution for ensuring safer and more organized public gatherings.

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